



Cite this: *Phys. Chem. Chem. Phys.*,
2020, 22, 5170

The puzzling hyper-fine structure and an accurate equilibrium geometry of succinic anhydride†

Michaela K. Jahn, ^a Daniel A. Obenchain, ^a K. P. Rajappan Nair, ^a
Jens-Uwe Grabow, ^a Natalja Vogt, ^{b,c} Jean Demaison, ^b Peter D. Godfrey ^d and
Don McNaughton ^{d,*}

An accurate semiexperimental equilibrium structure of succinic anhydride has been determined from a combination of experiment and theory. The cm-wave and mm-wave rotational spectra of succinic anhydride, 3,4-dihydrofuran-2,5-dione, were recorded in a pulsed supersonic jet using Fourier-transform microwave spectroscopy and in a free-jet using mm-wave absorption spectroscopy. Many lines in the cm-wave spectrum show fine structure and after eliminating all other possibilities the origin of this fine structure is determined to be from spin–spin interaction. Accurate rotational and quartic centrifugal distortion constants are determined. Assignments of ¹³C and ¹⁸O singly substituted isotopologues in natural abundance were used to obtain a substitution geometry for the heavy atoms of succinic anhydride. Theoretical approaches permitted the calculation of a Born–Oppenheimer *ab initio* structure and the determination of a semiexperimental equilibrium structure in which computed rovibrational corrections were utilized to convert vibrational ground state rotational constants into equilibrium constants. The agreement between the semiexperimental structure and the Born–Oppenheimer *ab initio* structure is excellent. Succinic anhydride has been shown to have a planar heavy atom equilibrium structure with the effects of a large amplitude vibration apparent in the resultant rotational constants.

Received 17th December 2019,
Accepted 12th February 2020

DOI: 10.1039/c9cp06775b

rsc.li/pccp

Introduction

Succinic acid is a dicarboxylic acid with four carbon atoms, occurring naturally in plant and animal tissues and it plays a significant role in intermediary metabolism (Krebs cycle). It is a colourless crystalline solid with a melting point of 118–120 °C and is soluble in water. Its anhydride, succinic anhydride, C₄H₄O₃, formally dihydro-2,5-furandione and shown in Fig. 1, is the simplest cyclic anhydride with a five-membered ring.

Succinic anhydride is of practical importance in the manufacture of polymeric materials (alkyd and other special resins) and is used in the manufacture of pharmaceuticals, agrochemicals, dyes, photographic chemicals, surface active agents, lubricant additives, organic flame retardant materials, esters, flavours and fragrances. Its applications include its use as a cross-linking agent in ion-exchange membranes, curing agent for epoxy resins and starch modifier in foods and feeds. An accurate determination of its molecular constants is important to determine its precise

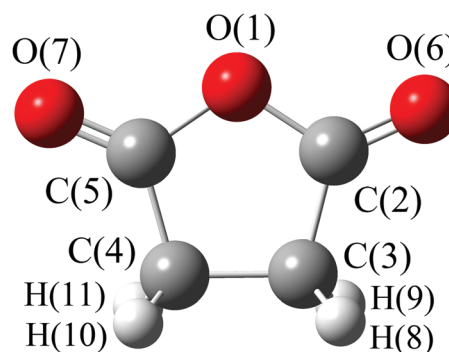


Fig. 1 Molecular model of succinic anhydride with atom numbering.

structure and hence a fundamental understanding of its chemical properties. Ehrenberg showed that succinic anhydride has an essentially planar heavy atom skeleton structure in the crystal form.¹ The first gas-phase study on succinic anhydride was carried out by Brendhaugen *et al.*² who studied the molecule by gas-phase electron diffraction (GED). The GED structure was recently reinvestigated by Vogt *et al.*³ considering the large amplitude ring-twisting motion. In that work, the equilibrium structure of succinic anhydride was derived from GED data taking into account non-linear kinematic, harmonic and anharmonic vibrational effects and computed by different

^a Institut für Physikalische Chemie & Elektrochemie,
Gottfried-Wilhelm-Leibniz-Universität, Germany

^b Section of Chemical Information Systems, University of Ulm, Germany

^c Chemistry Department, Lomonosov Moscow State University, Russia

^d School of Chemistry, Monash University, Wellington Rd, Clayton, Vic 3800,
Australia. E-mail: donald.mcnaughton@monash.edu

† Electronic supplementary information (ESI) available. See DOI: 10.1039/c9cp06775b

quantum-chemical methods (up to coupled-cluster level of theory). The C_{2v} point-group symmetry was assumed in the GED analysis³ following the prediction of coupled-cluster calculations.

The investigation of the pure rotational spectrum in the microwave region provides accurate values of the rotational constants and hence the molecular structure. The purpose of the present investigation was to determine an accurate molecular structure from microwave (MW) spectroscopy aided by computational calculations. A study of the ^{13}C and ^{18}O isotopologues has resulted in the determination of the structure of the heavy atom skeleton by Kraitichman's substitution method (r_s) which is compared with an *ab initio* equilibrium structure and a semiexperimental equilibrium structure. We have recently carried out similar MW spectroscopic studies on the related chemical systems through studies of fumaric acid supported by quantum chemical calculations⁴ and the equilibrium structure of maleic anhydride has recently been determined by the combined use of MW spectroscopy and *ab initio* methods.⁵ Spectral analysis was complicated because many of the transitions observed for succinic anhydride displayed partially resolved fine structure, which is in contrast to the spectrum recently observed by McMahon *et al.*⁶ They observed a simple doubling of all the rotational transitions of succinic anhydride and attributed it to quantum mechanical tunnelling between two pseudo-rotational states. We show by experiment and theory that this is not the case. The outcome of this high-resolution study supported by theoretical calculations is a precise equilibrium structure for succinic anhydride thereby exploring the planarity of the ring and also a determination of the origin of the observed fine structure.

Experimental

Rotational spectra in the mm-wave (48–87 GHz) were recorded in a Stark-modulated free-jet spectrometer previously described by Brown *et al.*⁷ In its current configuration the spectrometer utilizes solid state sweep oscillators extending the frequency range to approximately 100 GHz. Stark modulation was provided at 33 kHz between parallel-plate electrodes spaced ~ 3.5 cm apart. Succinic acid (99%, Sigma-Aldrich) and succinic anhydride (99%, Sigma-Aldrich) were vaporized at temperatures in the range 160–180 and 110–120 °C respectively, entrained in a stream of argon at a pressure of ~ 30 kPa and introduced between the Stark electrodes of the spectrometer *via* a 350 μm diameter pinhole nozzle held 10 °C above the vaporization temperature. Under these conditions, the post-expansion rotational temperature was *ca.* 10 K and in the case of succinic acid incomplete thermal decomposition to succinic anhydride was observed. Electric fields up to 1600 V cm^{-1} were used for Stark modulation. A Lorentzian line shape function was fitted to each resultant line profile resulting in typical full width half height (FWHH) for transitions of 100–200 kHz, leading to experimental line centre-frequency uncertainties of 10 to 40 kHz.

The spectrum in the cm-wave region was recorded on a Fourier-transform microwave (FT-MW) pulsed-supersonic jet expansion Fabry–Pérot-type resonator spectrometer in Hannover, using the coaxially oriented beam and resonator arrangement (COBRA).^{8–10} Extremely narrow line widths of < 5 kHz FWHH with a resolving power of < 500 Hz are achieved for unblended lines at reasonably high signal-to-noise ratio. Succinic anhydride was in the first case obtained by heating succinic acid (99%, Alfa Aesar) to ~ 170 °C, which decomposes almost completely to the anhydride under these conditions. Afterwards succinic anhydride (99%, Sigma-Aldrich) was used without further purification, was vaporized *in situ* employing a heated expansion nozzle, installed in the backside of one of the resonator reflectors. These measurements were carried out with a measured nozzle temperature ~ 60 °C with Neon (1–2 bar) used as carrier gas, resulting in rotational temperatures of *ca.* 2 K. All frequency measurements were referenced to a global positioning system (GPS) disciplined rubidium frequency standard.

Quantum-chemical calculations

Geometries, rotational constants, quartic centrifugal distortion constants, dipole moments, spin-rotation constants and harmonic vibrational frequencies of succinic anhydride were calculated using Gaussian 09.¹¹ Combinations of the B3LYP functional with the standard triple- ζ 6-311+G(d,p) basis set or the correlation-consistent polarized triple valence basis set (cc-pVTZ) of Dunning were used. MP2 calculations were carried out with cc-pVTZ, aug-cc-pVTZ and cc-pwCVQZ basis sets. Furthermore, the anharmonic force field was calculated up to semi-diagonal quartic terms at the B3LYP/cc-pVTZ level of theory. The rovibrational corrections to the ground-state rotational constants were calculated using quadratic and cubic force fields and were used to derive the semiexperimental equilibrium rotational constants. Finally, the structure was optimized at the level of coupled-cluster theory with a perturbative treatment of connected triples, CCSD(T), all electrons being correlated, AE, in conjunction with the cc-pwCVQZ basis set.

Results and discussion

Employing the wide frequency coverage and quite versatile sample vaporization capability of the mm-wave free jet spectrometer, many lines in excess to those attributable to succinic acid were observed when succinic acid was volatilized at its required temperature of *ca.* 180 °C. When the previously heated solid sample was reused it was found that the excess lines appeared at a much lower temperature of *ca.* 110 °C. Their suspected identity as the much more volatile anhydride was confirmed, firstly by their intense production when P_2O_5 was mixed with the succinic acid solid prior to heating at 110 °C, and finally when a sample of pure succinic anhydride was obtained and heated to 110 °C. The production of succinic anhydride from the acid varies in the two spectrometer systems used. By comparison with the mm-wave system, in the cm-wave

system extremely weak acid lines were observed on heating of the acid with almost complete conversion to the anhydride. In experiments on succinic acid at Universidad del País Vasco using UV laser ablation to volatilize the acid no anhydride lines were observed.¹² Similar observations had earlier been observed in the electron diffraction experiment of maleic acid where the substance decomposed into maleic anhydride.¹³ Maleic anhydride ($C_4H_2O_3$) and succinic anhydride ($C_4H_4O_3$) differ structurally by the CC double bond and the CC single bond between the methylene groups, respectively, but both have a molecular arrangement that leads to the elimination of water on heating. We surmise that the almost complete conversion to anhydride in the FTMW system is due to the longer retention time of the sample in the heater prior to the pulsed nozzle. Upon assignment of the main isotopologue of succinic anhydride, rotational constants for each of the ^{13}C and ^{18}O singly substituted isotopologues were estimated from MP2/cc-pVTZ level calculations, employing the systematic corrections found to be necessary between the calculated and experimental rotational constants of the main species. Spectra predicted using these corrected constants led to the detection and assignment of all of the possible singly substituted ^{13}C and ^{18}O isotopologues in natural abundance. In order to obtain more precise line positions and improved resultant constants, lines were subsequently measured at lower frequencies in the FT-MW system. Many of the lines measured with supersonic jet FT-MW spectroscopy in the lower frequency region involving low quantum numbers were found to be split into a number of components. The splittings are of the order of 10 kHz or less which are generally non observable in a typical broadband microwave spectrometer. Typical rotational transitions showing both fine structure and no fine structure are shown in Fig. 2. An initial assignment of the centre frequencies of the 57 transitions obtained from the FT-MW spectrometer in Hannover and 44 unsplit lines in the millimeter wave region of the main isotopologue were fitted by using Watson's S reduced Hamiltonian in the III^f representation¹⁴ in Pickett's SPFIT.¹⁵ The assigned lines are presented in ESI,[†] Table S1.

Lines of the singly substituted isotopologues ^{13}C and ^{18}O in natural abundance at all carbon and oxygen positions were measured in the cm region to complement those measured in the mm region. Many lines in the lower frequency region with low quantum numbers are again observed as split into complex multiplets as in the case of the parent molecule. The isotopologue lines were fitted using the mean unsplit frequency positions, which are given in Table S2 (ESI[†]). For the ^{18}O isotopologue where the atom is on the symmetry axis, two of the centrifugal distortion constants were held to those of the main species due to the smaller number of assigned transitions available. The resulting molecular constants for all the isotopologues are given in Table 1.

Structure

In agreement with the study of Vogt *et al.*,³ theoretical calculations at the DFT level show a planar heavy atom structure whilst the MP2/cc-pVTZ calculation results in a structure with the CH_2 groups twisted to give a non-planar heavy atom structure.

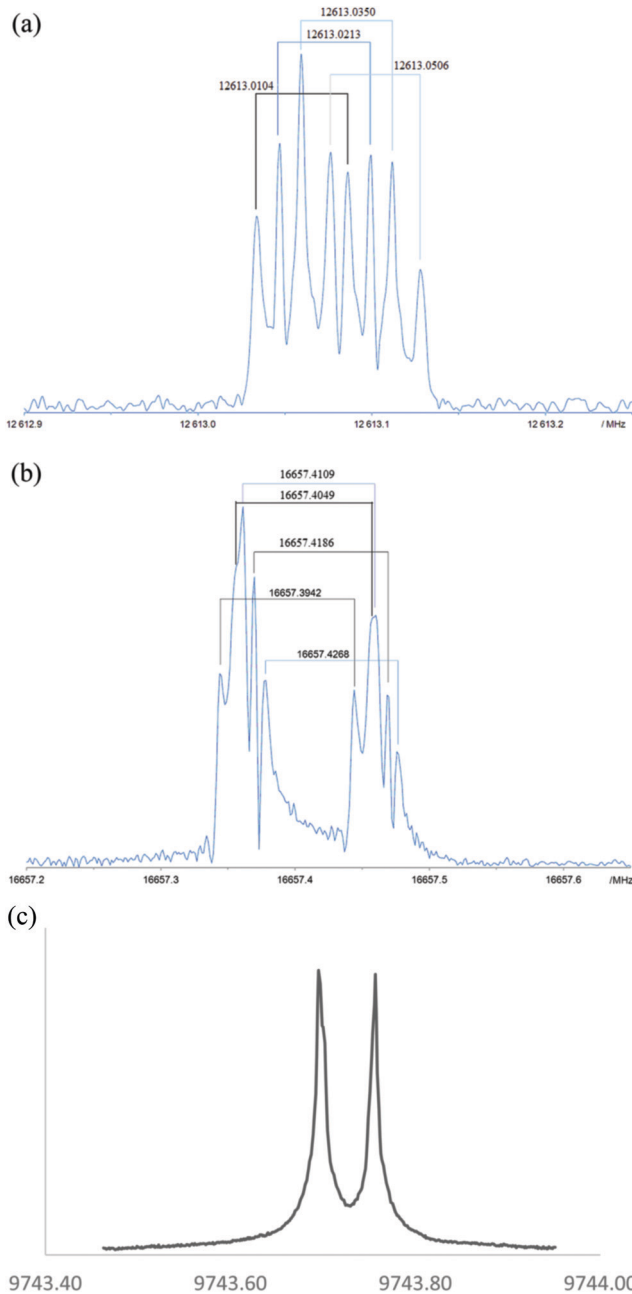


Fig. 2 (a) Hyperfine structure in $2_{2,1}-2_{1,2}$ rotational transition in succinic anhydride, (b) hyperfine structure in rotational transition $5_{2,4}-5_{1,5}$ in succinic anhydride. (c) $4_{2,2}-4_{1,3}$ rotational transition in succinic anhydride showing no resolved hyperfine splitting.

1. Planar moments of inertia

Experimentally derived planar moments of inertia can also be calculated to explore the planarity of the molecule as in Bohn *et al.*¹⁶ Bohn showed for a number of molecules that the hydrogens of each CH_2/CH_3 group contributes $1.56 \text{ u}\text{\AA}^2$ to the planar moment of inertia. If the heavy-atom skeleton is planar, the planar moment of inertia P_{cc} may be written:

$$P_{cc} = 2m_H d_{HH}^2 = -\Delta/2 \quad (1)$$

Table 1 Rotational and centrifugal distortion constants Watson's S reduction in the III' representation of isotopologues of succinic anhydride/MHz. Numbers in parentheses are 1σ uncertainties in the last digit of the constant. Constants without uncertainties are fixed at the values of the parent

Molecular constants	Parent	^{13}CH isotope	^{13}CO isotope	^{18}OCO isotope	$^{18}\text{OCOC}$ isotope
A_0	5932.643730(84)	5815.68027(34)	5931.40087(34)	5881.70709(129)	5816.02216(106)
B_0	2363.272417(40)	2356.839147(190)	2349.061863(183)	2259.36835(40)	2363.40737(38)
C_0	1728.290590(41)	1714.846042(75)	1720.573515(75)	1667.987119(98)	1718.361497(126)
D_J	0.00010085(33)	0.00010082(164)	0.00010029(150)	0.00009279(247)	0.00010085
D_{JK}	0.00050601(104)	0.0004867(281)	0.0004719(281)	0.000537(207)	0.00050601
D_K	0.0013770(51)	0.001429(75)	0.001418(75)	0.001417(38)	0.0017551(217)
d_1	−0.000028155(40)	−0.00002954(133)	−0.00002922(123)	−0.0002574(189)	−0.00002133(60)
d_2	−0.0000033788(193)	−0.00000449(102)	−0.00000361(100)	−0.00000403(74)	−0.00000424(60)

Table 2 Planar moments of inertia of succinic anhydride (P_{cc}) $\mu\text{Å}^2$

$^{12}\text{C}^{16}\text{O}$	$^{13}\text{C}_{3,4}$	$^{13}\text{C}_{2,5}$	$^{18}\text{O}_1$	$^{18}\text{O}_{6,7}$	$^{12}\text{C}^{16}\text{O}$ B3LYP	$^{12}\text{C}^{16}\text{O}$ MP2(twist)
3.308652(5)	3.311065(11)	3.308853(22)	3.311966(22)	3.309167(1)	3.091876	3.479972

m_{H} is the mass of the out-of-plane hydrogen atoms, d_{HH} the non-bonded distance between the two out-of-plane hydrogen atoms, and Δ is termed the pseudo-inertial defect. For a planar heavy atom structure for succinic anhydride a value of 3.12 uÅ^2 is expected, and this is close to the value of 3.09 uÅ^2 predicted by theory for the DFT equilibrium structure as shown in Table 2. A non-planarity of the heavy-atom skeleton would also lead to a significant isotopic dependence of P_{cc} which from the values in Table 2 is not observed. It is established that the ground state moment of inertia is not a good measure of planarity because the vibrational contribution can be extremely large.¹⁷ However, using the anharmonic force field it is possible to calculate the rovibrational contributions to the ground state rotational constants, allowing us to deduce the equilibrium value of P_{cc} giving $d_{\text{HH}} = 1.7576(1) \text{ Å}$ (mean of five values) in perfect agreement with the equilibrium structure presented in the next section.

Using the semiexperimental rotational constants, it is possible to calculate the vibrational contribution to the pseudo-inertial defect, $\Delta_v = -0.391 \text{ uÅ}^2$. Such large negative values are mainly a consequence of a low-frequency out-of-plane vibration. Succinic anhydride has indeed a low frequency large amplitude motion of the oxygen atoms out of the heavy atom plane which will also contribute to the planar moment of inertia as outlined by Oka¹⁸ and shown by us previously to be valid for the ground state constants of planar polyaromatic species.¹⁹ The difference between the predicted and measured planar moment of inertia is consistent with that due to a low frequency vibration much less than 100 cm^{-1} , which is compatible with the values of

$60\text{--}80 \text{ cm}^{-1}$ predicted from the theoretical calculations. It is possible to confirm this result using the force field. It gives $\Delta_v = -0.395 \text{ uÅ}^2$, in good agreement with the experimental value.

For the MP2 twisted structure, the calculated value of the planar moment in Table 2 is 10% larger than that from experiment. Hence the measured planar moment of inertia is compatible with a planar heavy atom ground state. For the twisted structure the energy difference between the minima and the saddle point is extremely small with an extremely shallow potential, thus both B3LYP and MP2 calculations disclose the planar and quasi-planar equilibrium configurations of the skeleton, respectively, and are compatible with an essentially planar skeleton for the vibrational ground state structure. Moreover, the MP2 method sometimes fails to give a completely correct structure because, in some cases, it is not able to compute the torsional angles with an acceptable accuracy. A particularly relevant example is given by formamide, and it's rich history is reviewed in Fogarasi and Szalay.²⁰

2. Substitution structure

Substitution coordinates of the heavy atoms were calculated with and without assumptions on the symmetry of the molecule. For the no symmetry case the c coordinates, when not imaginary, are not significantly different from zero. However, this is not enough to warrant that the heavy-atom skeleton is planar. Furthermore, several coordinates are small and, as a consequence, they are probably not accurate. This is confirmed by calculating the coordinates from the semiexperimental equilibrium rotational constants, which are included in Table 3

Table 3 Substitution structure r_s : Cartesian coordinates in the principal axis system (all values in Å)

Substitution position	From ground state rotational constants			From semiexperimental equilibrium rotational constants		
	$ a $	$ b $	$ c $	$ a $	$ b $	$ c $
C(3,4) ^{13}CH	0.7600(20)	1.3154(11)	0.048(32)	0.7598	1.3143	0.0229
C(2,5) ^{13}CO	1.1411(13)	0.1348(111)	1^a	1.1405	0.1544	1^b
O(1) ^{18}OCO	2.2308(7)	0.6357(24)	0.009(166)	2.2251	0.6349	0.017
O(6,7) $^{18}\text{OCOC}$	1^a	0.9315(16)	0.040(37)	0.0203	0.9286	0.027

^a Imaginary value. ^b Imaginary value ≈ 0 .

Table 4 *Ab initio* structure calculated at the CCSD(T)_AE level of theory (bond lengths in Å and angles in degrees)

Basis set	wCVTZ [3]	wCVQZ	CBS ^a
C–O(1)	1.3831	1.3809	1.3792
C–O(1)–C	111.25	111.37	111.46
=C(2)–C(3)	1.5155	1.5134	1.5119
O(1)–C–C	109.86	109.81	109.78
C–O(1)–C–C	–1.1373	–0.3132	0.2882
C–H(9)	1.0873	1.0870	1.0868
C–C–H(9)	108.65	108.41	108.23
O(1)–C–C–H(9)	125.07	122.66	120.91
C–H(8)	1.0882	1.0873	1.0866
C–C–H(8)	108.09	108.25	108.36
O(1)–C–C–H(8)	–118.30	–120.79	–122.61
C(2)=O	1.1900	1.1878	1.1861
O=C–O(1)	121.61	121.59	121.58
C–O(1)–C=O	–180.86	–180.23	–179.77

^a Complete basis set wCV ∞ Z, see text.

along with the substitution coordinates not assuming planarity of the ring and C_{2v} symmetry.

3. Born–Oppenheimer equilibrium structure

The structure of succinic anhydride was optimized assuming a non-planar C_2 symmetry at the CCSD(T)_AE level of theory using the cc-pwCVTZ and cc-pwCVQZ basis sets with results presented in Table 4. With the cc-pwCVTZ basis set, the predicted structure of the heavy-atom skeleton is significantly non-planar but with the cc-pwCVQZ basis the prediction becomes much closer to planar with C_{2v} symmetry. With the basis sets chosen CCSD(T) does not yield planar equilibrium geometries but as the basis gets larger the nonplanarity decreases. A corroboration is obtained by extrapolating the structural parameters to the complete basis set limit using the following empirical formula:

$$r(n) = r_{\infty} + An^{-3}; n = 3, 4 \text{ (i.e. } T, Q) \quad (2)$$

This approximate formula²¹ is known to give satisfactory results for first-row atoms although it slightly exaggerates the correction. The results are given in Table 4. The coordinates of the hydrogen atoms are now almost identical and the dihedral angles involving the atoms of the heavy skeleton are close to zero or 180° and confirm that the extrapolation somewhat exaggerates the correction.

4. Semiexperimental equilibrium structure

Semiexperimental equilibrium rotational constants were calculated from the experimental ground state rotational constants and the rovibrational corrections computed at the B3LYP/cc-pVTZ level of theory and are given in ESI,† Table S3. It is possible to estimate the uncertainty of these semi-experimental constants because the equilibrium planar moment of inertia P_c should be constant for the parent species and the isotopologues. Their small variations may be used to estimate the standard deviation of P_c which is approximately $0.0010 \mu\text{Å}^2$. Using the law of propagation of errors, the standard deviation may also be written

$$\sigma^2(P_c) = (\sigma^2(I_a) + \sigma^2(I_b) + \sigma^2(I_c))/4 \quad (3)$$

Table 5 Structure of anhydride succinic anhydride (bond lengths in Å and angles in degrees)

	r_s^a	r_c^{SE}	r_c^{BO}	CCSD(T)_AE/wCV ∞ Z ^d	r_c^{SEc}
		GED ^b			^e
C–O	1.3917(65) ^c	1.3833(27)	1.3792	1.3824(5)	1.3824(5)
C–O–C	110.16(40)	110.88(25)	111.46	111.32(3)	111.32(3)
C=O	1.1993(49)	1.1881(10)	1.1861	1.1878(5)	1.1878(5)
O=C–O	120.39(13)	121.46(17)	121.58	121.52(3)	121.52(3)
=C–C	1.4994(5)	1.5142(10)	1.5119	1.5116(3)	1.5116(3)
O–C–C	110.20(16)	110.13(16)	109.78	109.77(5)	109.77(5)
C–H		1.0849(61)	1.0867	1.090(2)	1.090(2)
=C–C–H		106.86(108)	108.29	108.34(12)	108.34(11)
O–C–C–H		121.8 ^f	121.76	121.83(21)	121.84(20)

^a Assuming C_{2v} symmetry. ^b Large-amplitude motion model. Parenthesized uncertainties are $2.5\sigma_{LS}$ including a systematic error. ^c The uncertainties are 1σ . ^d Fit with three predicates (mean of CCSD(T)_AE/wCVQZ values): C–H = 1.0872(20) Å, C–C–H = 108.32(30)°, O–C–C–H = 121.73(30)°. ^e Fit with an additional fourth predicate: C–C = 1.512(2) Å (CBS value). ^f Assumed at the r_c^{BO} value, see ref. 3.

We may assume that $\sigma(I_b) = \sigma(I_c)$ (both I_b and I_c have similar values and similar rovibrational corrections). Furthermore, I_a is approximately 2.5 times smaller than I_b , thus we may assume that $\sigma(I_a) \approx \sigma^2(I_b)/2.5$. In conclusion $\sigma^2(P_b) \approx 2.3\sigma(I_b)$. Hence, $\sigma(I_b) \approx \sigma^2(I_c) \approx 0.0004 \mu\text{Å}^2$, i.e. an uncertainty of about 0.005 MHz for B_c . These values are used for a first estimate of the weights in the least-squares fit.

The available rotational constants do not permit to determine a complete structure because the hydrogen atoms were not substituted. However, the CH bond lengths and $\angle$ CCH bond angles can be obtained extremely accurately at the CCSD(T)_AE/cc-pwCVQZ level of theory. These *ab initio* values can be used as predicates in the least-squares fit²² with conservative uncertainties (to estimate their weight) of 0.002 Å for the CH bond lengths and 0.3° for the bond angles. The semiexperimental structure is given in Table 5 where it is compared with the other structures. The agreement between the semiexperimental structure and the Born–Oppenheimer *ab initio* structure is good. The r_c^{SE} bond lengths determined from data of two different methods (GED³ and MW) agree well with each other (within their uncertainties). The differences between the bond angles are only a few tenths of a degree outside the estimated uncertainties of the GED values.

Hyperfine structure

Unlike the recent work of McMahon *et al.*⁶ we find not a simple, well-spaced doubling of all the lines, but some single lines (Doppler doubled), and some lines with fine structure below the resolution of their instrument and not assignable to pseudo rotational states. Fig. 2 shows three examples of observed lines. For inversion states 0^+ and 0^- as invoked by McMahon *et al.* one would expect a simple doubled set of lines of approximately equal intensity and almost constant spacing from line to line. This is not what is observed by McMahon *et al.*⁶ and their observed spectrum is most likely due to their instrumental

Table 6 Spin-rotation constants

	Ketene DFT ^a	Ketene exp. ^b	Methane DFT	Methane exp. ^b	Succinic anhydride DFT	Succinic anhydride exp. ^c	Oxirane DFT
C_{aa} kHz	−19.21	−18.296	−11.16	−10.372	−0.568	−10.33	−1.34
C_{bb} kHz		−0.087			−0.163	−1.03	−2.32
C_{cc} kHz	−1.21	−1.156					−0.274

^a B3LYP/cc-pVTZ. ^b From ref. 23. ^c Derived from fitting observed gross fine structure as spin-rotation hyperfine.

conditions and not quantum mechanical tunnelling. The other possible explanations for the fine structure observed in our study are spin-rotation or spin-spin coupling of four equivalent protons (or two sets of equivalent protons for a non-planar skeleton). The hyperfine structure present can be approximately fitted with four equivalent spin-rotation constants implying C_{2v} symmetry for the ground state of succinic anhydride. However, the resultant experimental hyperfine constants (presented in Table 6) are more than an order of magnitude larger than those predicted by B3LYP/cc-pVTZ calculations (also included in Table 6) which would result in line splittings well below the resolution of the COBRA spectrometer. Using *ab initio* methods or different basis sets does not markedly change the predicted values of the spin-rotation constants. Benchmark calculations on other molecules, included in Table 6, show the B3LYP/cc-pVTZ calculation of spin-rotation constants to be reliable to within 10%. The observed splittings are thus not due to spin-rotation.

The spin-spin splitting constants can be calculated purely from the molecular geometry. Spin-spin splitting constants of hydrogens separated by 3 bond lengths should also give fine structure well below the resolution of the spectrometer. This leaves the spin-spin splitting of hydrogen atoms bonded to the same carbon atoms or adjacent carbon atoms and these can be calculated from eqn (4) using *ab initio* structural parameters.

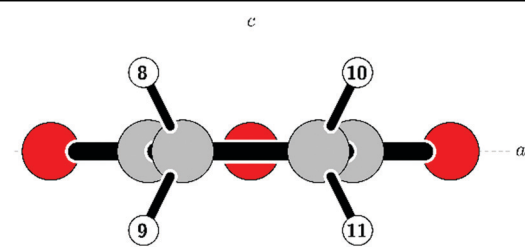
Using structures optimized at the MP2/aug-cc-pVTZ level of theory, where both planar and non-planar heavy atom structures are available, the r_{H-H} and the $\mathbf{r}_i$ angle (the angle between the H-H and the principal axis) were taken from the resulting structure files. These values were used in the following equation to predict each of the nuclear spin-nuclear spin components, D_{ii} , where i represents each of the principal axis components, a , b , and c while g_H are the nuclear g tensors for the H nuclei:

$$D_{ii} = -7.70000 \text{ kHz } \text{\AA}^3 \times \frac{g_{H1}g_{H2}}{r^3} \times \frac{1}{2}(3 \cos^2(\angle \mathbf{r}_i) - 1) \quad (4)$$

eqn (4) is taken from Grubbs *et al.*²⁴ and was adapted from that originally used to describe the spin-spin interaction by Read and Flygare²⁵ in the acetylene-HF complex. Tables 7 and 8 contain the calculated nuclear spin-nuclear spin coupling terms for the planar and twisted heavy atom conformations. Because the D tensor is traceless, only the two components used in Pickett's programs¹⁵ are given.

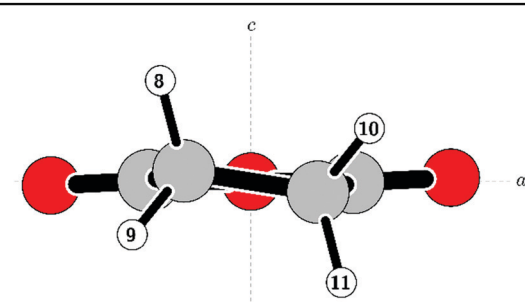
Fig. 3 shows the predicted and experimental line profiles for the $3_{22}-3_{13}$ transition using only the largest coupling, that between the two pairs of geminal protons for the planar structure. As can be seen the comparison is quite good and

Table 7 Predicted nuclear spin-nuclear spin coupling terms for the planar conformation. Based on the structure optimized at the MP2/aug-cc-pVTZ level of theory. As the D tensor is traceless, only the two components used in Pickett's programs are given



Paired hydrogen atoms	Coupling term	Value (kHz)
8–9	$3/2 \cdot D_{aa}$	33.45
10–11	$1/4 \cdot (D_{bb} - D_{cc})$	16.73
8–11	$3/2 \cdot D_{aa}$	−6.58
9–10	$1/4 \cdot (D_{bb} - D_{cc})$	1.21
8–10	$3/2 \cdot D_{aa}$	−26.46
9–11	$1/4 \cdot (D_{bb} - D_{cc})$	0.00

Table 8 Predicted nuclear spin-nuclear spin coupling terms for the twist conformation. Based on the structure optimized at the MP2/aug-cc-pVTZ level of theory. As the D tensor is traceless, only the two components used in Pickett's programs are given



Paired hydrogen atoms	Coupling term	Value (kHz)
8–9	$3/2 \cdot D_{aa}$	30.11
10–11	$1/4 \cdot (D_{bb} - D_{cc})$	15.43
8–11	$3/2 \cdot D_{aa}$	−2.28
9–10	$1/4 \cdot (D_{bb} - D_{cc})$	1.82
8–10	$3/2 \cdot D_{aa}$	−23.64
9–11	$1/4 \cdot (D_{bb} - D_{cc})$	0.24

the calculated spin-spin splitting components are a close match to line profiles for both split and unsplit lines. The observed structure consists of many overlapped spin-spin hyperfine components as well as the much smaller spin-rotation splitting so at present the hyperfine constants cannot be directly derived from the present data. It is apparent though that the gross

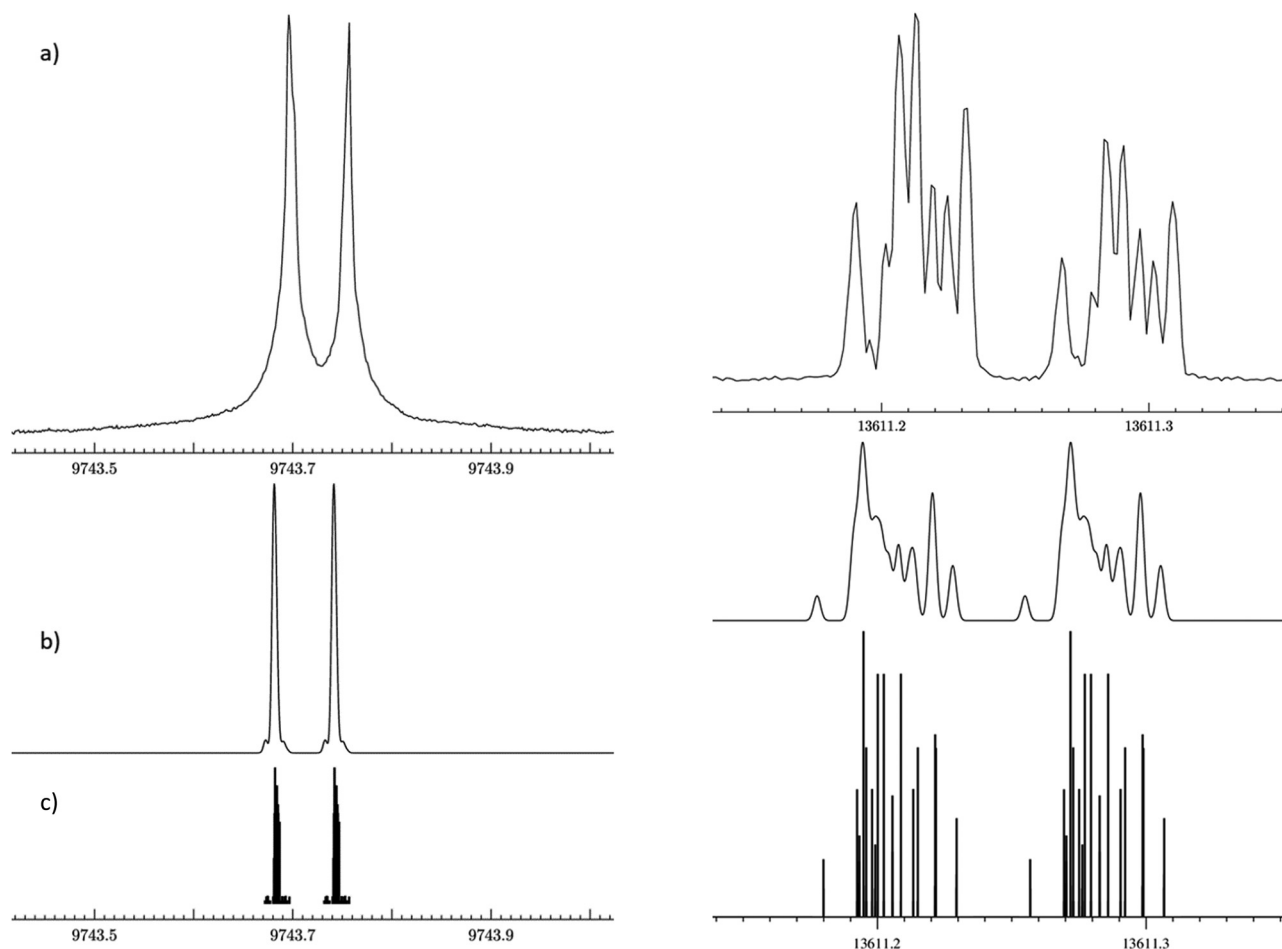


Fig. 3 (a) Experimental and (b and c) spin-spin predicted transitions $4_{2,2} \leftarrow 4_{1,3}$ and $3_{2,2} \leftarrow 3_{1,3}$.

hyperfine structure observed is due to spin-spin splitting. Given that the predicted values for the spin-spin hyperfine components differ by around 10% for planar/non-planar heavy atom skeletons, measurement of these parameters and comparison with theory could provide a method to determine the planarity or non-planarity of the heavy atom system.

Conclusion

An accurate equilibrium structure of succinic anhydride has been determined from a combination of jet cooled rotational spectroscopy of the isotopologues of the heavy atoms and computation. Many lines in the cm-wave spectrum show fine structure and the origin of this fine structure is determined to be from spin-spin interaction. Theoretical approaches permitted the calculation of a Born-Oppenheimer *ab initio* structure (r_e^{BO}) and the determination of a semiexperimental equilibrium structure (r_e^{SE}) where computed rovibrational corrections were utilized to convert vibrational ground state rotational constants into equilibrium ones. The r_e^{BO} structure is found to be in excellent agreement with the r_e^{SE} structure. Succinic anhydride has been shown to have a planar heavy

atom equilibrium structure with the effects of a large amplitude vibration apparent in the resultant rotational constants. Contrary to the spectroscopic method, the structure determination from GED data requires some geometrical assumptions (for instance for difference between two slightly different C-C bond lengths). Moreover, the GED structure has larger experimental uncertainties. Similar to succinimides,^{26,27} the planarity of the heavy-atom skeleton can be explained by the influence of two carbonyl groups with the C_{sp^2} atoms which require enlarging of the adjusted ring angles and thus lead to stabilization of a planar ring conformation in spite of the repelling of two methylene groups.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We gratefully acknowledge support from the Deutsche Forschungsgemeinschaft (DFG), the Land Niedersachsen, and the Dr Barbara-Mez-Starck Stiftung. DAO would like to thank the Alexander von Humboldt Foundation for funding.

The Hannover group acknowledges support from the Leibniz Universität IT Services (LUIS) cluster.

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